FOR COMPLETE BEGINNERS

# $120 basket. No gift?

## Complete Modern Java, episode 11: and, or, not and else if, in Java and TypeScript

> Combine conditions with && || !, and choose one of several outcomes with else if. The first true condition decides.

Gift from $100. Free delivery from $50, members only. Everyone else $4.99.

Two things at once, and one of three outcomes.

### In this episode

- && and || and !
- else if chains
- Why branch order matters
- Short circuit
- Brackets with && and ||

### Coming next

- Loops (episode 12)
- Methods (episode 13)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

## 01 What each line printed

| | |
|---|---|
| 5240 >= 5000 && member | `true` |
| 5240 >= 10000 \|\| member | `true` |
| !member (true) | `false` |
| tiers: 5240 member / not member | `Free delivery / Delivery: $4.99` |
| tiers: 12000 | `Free gift` |
| wrong order: 12000 and a member | `Free delivery (gift skipped)` |

Every row was run, not predicted.

## 02 else if

Java checks the conditions in order. The first true one decides; the rest are not checked.

Put the most specific rule first. && stops at the first false, so a guard on its left protects the right side.

Tiers.java

```
if (totalCents >= 10000) {
    IO.println("Free gift");
} else if (totalCents >= 5000 && member) {
    IO.println("Free delivery");
} else {
    IO.println("Delivery: $4.99");
}
```

> Printed Free delivery, Delivery: $4.99 and Free gift for the three cases.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

## 03 A guard with &&

items is 0, so items > 0 is false and the division is never evaluated: no ArithmeticException.

Average.java

```
if (items > 0 && totalCents / items > 1000) {
    IO.println("Average above $10");
} else {
    IO.println("No qualifying average");
}
```

> Printed No qualifying average, with no error.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

## Write rules that combine

1. Write the rules in words, one outcome per customer.
2. && for both, || for either, ! to flip.
3. Most specific rule first in an else if chain.
4. Guard a risky right side with && on the left.
5. Mixing && and ||? Add brackets.

### Everything else

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**The code and the challenge files: complete-modern-java**
https://github.com/code-with-sam-dev/complete-modern-java

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